

Veridicus Pre-Registered Prediction Card

For: Staff Scientist, Recursion Pharmaceuticals | Target: Campaign X (neurodrug program)

Predicted lift (committed BEFORE customer data arrives):

Metric	Baseline B	Memory-augmented M	Delta (M - B)
Top-1 direction-hit rate	50.0%	75.7% (point)	+25.7 pp [+18.6, +29.3] pp
Cycle-waste delta (K=3)	n/a	n/a	-15.9 pp [-18.8, -10.2] pp (supporting)

Derivation formula (verbatim):

$\Delta_{\text{predicted}} = \alpha * \Delta_{\text{chembl}} + (1 - \alpha) * (\Delta_{\text{medauth}} * \tau)$

α in [0.5, 0.8] (weight toward in-domain ChEMBL evidence); point estimate $\alpha = 0.70$. τ in [0.3, 0.7] (cross-domain transfer degradation on the medical-auth prior); point estimate $\tau = 0.50$. $\Delta_{\text{medauth}} = 4.9\text{pp}$ (v0.5 medical-auth benchmark: 93.5% → 98.4% on 50 train → 148 held-out, 3 seeds $\pm 0.4\%$). $\Delta_{\text{chembl}} = 35.7\text{pp}$ observed on a 4-DP ChEMBL-grounded dry-run (scaffolding; not a customer-deliverable result); 90% bootstrap CI on the lift [+21.4, +50.0]pp across 7 seeds. Interval is constructed by sweeping α and τ across the four corner cases, taking min/max, and capping the full interval width at 15pp (+/- 7.5pp) to remain falsifiable without being vacuous.

Scoring criterion:

The prediction succeeds if, on your de-identified campaign data with decision points marked, the observed memory-augmented top-1 direction-hit rate minus baseline falls within [+18.6, +29.3] pp. Outside that interval in either direction = miss; the miss direction and magnitude are reported honestly. Cycle-waste delta is reported alongside but does not affect pass/fail.

Pre-commitment mechanism:

This PDF is committed to the Veridicus git repository. git commit hash: dfca94087e24406ff701daa685bee9d178fb9f96. The commit is timestamped via OpenTimestamps (bitcoin-anchored public timestamp); the .ots receipt accompanying this PDF is the falsifiable pre-commitment artifact. If the .ots receipt is in 'pending' state at delivery time (OTS attestations take ~1-6 hours to finalize), the finalized receipt will be forwarded post-hoc; the commit-hash alone is sufficient for third-party verification.

Scope & caveats:

(1) The generalization claim rests on the metacognitive rule proposer path (v0.5 medical-auth 93.5% → 98.4% on 148 held-out, 3 seeds $\pm 0.4\%$). A separate failure-driven extraction path exists in the production system but is intentionally EXCLUDED from this retrospective because Experiment E (LOO-validated) showed it memorizes rather than generalizes. (2) The ChEMBL dry-run is scaffolding, not a published number; it validates the harness and the +3pp lift gate but does not constitute a drug-discovery generalization result. (3) Cross-domain transfer from medical-auth to pharma lead-op reasoning is an open empirical question — the τ parameter in the formula models this explicitly. (4) Top-1 direction-hit is the headline metric; cycle-waste delta is supporting and may be withdrawn if noisy on your data.

Harness gate check: Lift 35.71pp with 90% CI [21.43, 50.00]pp on 4 DPs. GATE PASSED.